MIL-T-36888
5 September 1972

MILITARY SPECIFICATION

TEST KIT, NITROGEN DETERMINATION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers a Test Kit, Nitrogen Determination, suitable for use in the colorimetric or spectrophotometric determination of urea nitrogen, pre-formed ammonia, and non-protein nitrogen. Potency: 24 months.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

PPP-C-00186a Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specification and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
General Regulations for its Enforcement.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20402.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a test kit containing sufficient quantities of the four (4) necessary reagents, together with specific instructions for the intended determinations in clinical specimens of serum, plasma, or urine. The kit shall consist of the following:

(a) Urease, dried reagent: One (1) bottle of 20 ml size.
(b) Phenol color reagent, concentrated: One (1) bottle of 5 grams.
(c) Alkali-hypochlorite reagent, concentrated: One (1) bottle of 14 ml.
(d) Nitrogen standard: One (1) bottle of 3.5 ml size.

3.1.1 Characteristics of each reagent.

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3.1.1.1 Urease, dried reagent. Shall contain urease (activity at least approximately 1000 units per gram), 30 units; disodium edetate, 0.20 gram; sodium azide, 0.02 gram. The dried reagent shall be a white or slightly tan colored porous solid which shall give a clear, practically colorless working solution after being reconstituted for use with 20 ml of Purified Water, U.S.P. In addition, the urease, dried reagent shall comply with the following:

3.1.1.1.1 Net weight. Net weight of the contents of the bottle shall be not less than 190 mg and not more than 235 mg.

3.1.1.1.2 Urease activity assay. Shall assay not less than 0.2 and not more than 0.6 mEq NH_4^+/minute/bottle at 37° C., when determined as described in 4.5.1.1.

3.1.1.1.3 Limit of free ammonia. Shall contain less than 0.0025 mEq NH_4^+/bottle, when determined as described in 4.5.1.2.

3.1.1.2 Phenol color reagent. Shall contain sodium nitroferricyanide (nitroprusside), 0.025 gram; phenol U.S.P. melted crystals, 5 ml. The solid reagent shall be a pinkish or brownish crystalline mass showing small reddish, brownish, or black crystals (of nitroprusside) in or under it which shall give a clear, practically colorless working solution after being diluted to a total volume of 100 ml with Purified Water, U.S.P. In addition, shall comply with the following:

3.1.1.2.1 Net contents. Net contents of bottle shall weigh not less than 4.9 grams and not more than 5.7 grams, when determined by difference.

3.1.1.2.2 Ammonia limit. No blue color shall develop and solution shall show an absorbance less than 0.060 when determined as described in 4.5.2.1.

3.1.1.2.3 Test for correct color development. Absorbance of solution shall be not less than 0.450 and not more than 0.550 when determined as described in 4.5.2.2.

3.1.1.3 Alkali-hypochlorite reagent. Shall contain 4 ml of approximately 5.25 percent sodium hypochlorite; sodium hydroxide, 10 ml of a solution containing 25 grams NaOH per 100 ml. The concentrated reagent shall be clear, colorless or slightly yellow in color and shall give a clear, colorless working solution after being diluted to a total volume of 100 ml with Purified Water, U.S.P. In addition, shall comply with the following:

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3.1.1.3.1 Assay for alkalinity. Shall consume not less than 0.40 ml and not more than 0.52 ml of 1N Hydrochloric Acid Volumetric Solution, when assayed as described in 4.5.3.1.

3.1.1.3.2 Assay for hypochlorite content. Shall consume not less than 0.20 ml and not more than 0.45 ml of 1N Sodium Thiosulfate Volumetric Solution when assayed as described in 4.5.3.2.

3.1.1.4 Nitrogen standard. Shall contain not less than 3.5 ml of a sterile, clear Ammonium Sulfate, A.C.S. aqueous solution (15 mg N/100 ml). In addition, shall comply with the following:

3.1.1.4.1 pH. Shall be not less than 1.50 and not more than 2.50, when determined potentiometrically at 25° C., using the U.S.P. method.

3.1.1.4.2 Assay. Shall contain not less than 14.5 mg and not more than 15.5 mg of nitrogen per 100 ml when determined as specified in 4.5.4.

3.1.1.4.3 Sterility. Shall comply with the requirements and tests of the U.S.P. for sterility, or as an alternate, those of the Bureau of Biologics.

3.2 Ingredients. All ingredients used in the manufacture of the reagents shall be of U.S.P. or N.F. quality or, if not included in either of these compendia, the ingredients shall be suitable for their intended use.

3.3 Delivery. Not less than 21 months of the expiration dating period (potency period) shall remain at time of delivery to the Government. The expiration date on the product shall be based upon the shortest dated component in the test kit which shall be not less than 24 months.

3.4 Workmanship. Workmanship shall be first class throughout. The material and its containers shall be free from any defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of reagents, produced from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.3 Inspection. Inspection shall be conducted, as necessary, to determine compliance with the requirements of this specification.

4.4 Sampling. Sampling shall be conducted in accordance with the procedures set forth in MIL-STD-105, with an acceptable quality level (AQL) of 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

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4.5 Tests.

4.5.1 Urease dried reagent.

4.5.1.1 Urease activity.

Reagents:

(1) Purified Water, U.S.P.

(2) Phenol-catalyst reagent reconstituted to 100 ml with Reagent No. (1)

(3) Alkali-hypochlorite reagent reconstituted to 100 ml with Reagent No. (1)

(4) 1 N hydrochloride acid.

(5) Substrate: 1 percent urea in 1 percent disodium edetate adjusted to pH 6.5.

Reconstitute bottle of urease dried reagent with 20.0 ml Reagent No. 1.

Procedure:

Add 0.10 ml of sample to a test tube and warm to 37° C. Add 1.0 ml of substrate (prewarmed to 37° C.) and start stopwatch. Incubate for exactly 5.0 minutes. Stop reaction with 1.0 ml of 1 N hydrochloric acid.

Transfer 0.10 ml to 1.0 ml of phenol-catalyst reagent. Then add 1.0 ml of alkali-hypochlorite solution. Develop color at 37° C. for 15 minutes. Add 7.0 ml of Purified Water, U.S.P. Determine absorbances of sample solution and that of the standard solution* versus a reagent blank** with a suitable spectrophotometer in 1-cm cell, at 640 mu.

Prepare suitable sample and standard blanks substituting equivalent amounts of Purified Water, U.S.P. for the phenol catalyst solution and alkali-hypochlorite solution.

Calculation:

$$\frac{\text{Sample - Blank}}{\text{Standard - Blank}} \times 0.000214 \times \frac{20}{0.1} \times \frac{2.1}{0.1} \times \frac{1}{5} = \text{mEq } NH_4^+/\text{min/bottle}$$

$$\frac{\text{Sample - Blank}}{\text{Standard - Blank}} \times 0.18 = \text{mEq } NH_4^+/\text{min/bottle}$$

*For standard solution, substitute 0.020 ml of 10.7 mEq ammonium reference standard (15 mg N/100 ml) for the 0.1 ml of sample.

**For reagent blank, substitute 0.1 ml of Purified Water, U.S.P. for sample.

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4.5.1.2 Limit free ammonia. Add 0.5 ml of reconstituted urease to 1.0 ml of phenol catalyst reagent and 1.0 ml of alkali-hypochlorite solution. Incubate at 37° C. for 15 minutes. Add 7.0 ml of Purified Water, U.S.P. Determine absorbances of sample solution and that of standard solution versus a reagent blank with a suitable spectrophotometer in a 1-cm cell at 640 mu. (For reagent blank, use Purified Water, U.S.P. in place of sample, and for standard, in lieu of 0.5 ml of reconstituted urease, use 0.5 ml of Purified Water, U.S.P. plus 0.010 ml of a 10.7 mEq (15 mg N/100 ml) ammonia sulfate reference standard.

Prepare suitable sample and standard blanks substituting equivalent amounts of Purified Water, U.S.P. for the phenol catalyst solution and alkali-hypochlorite solution.

Calculation:

$$\frac{\text{Sample-Blank}}{\text{Standard Blank}} \times 0.000214 \times \frac{20}{0.5} = \text{mEq } NH_4^+/\text{bottle}$$

$$\frac{\text{Sample-Blank}}{\text{Standard-Blank}} \times 0.0086 = \text{mEq } NH_4^+/\text{bottle}$$

4.5.2 Phenol color reagent.

4.5.2.1 Ammonia limit. Dissolve contents of bottle with Purified Water, U.S.P. and dilute to 100.0 ml. To 1.0 ml of the diluted reagent, add 1.0 ml of diluted ammonia-free alkali-hypochlorite reagent. Add 7.0 ml of Purified Water, U.S.P. Incubate at 37° C. for 15 minutes, observe for the presence of color. Determine the absorbance of the solution at 640 mu with a suitable spectrophotometer in a 1-cm cell versus Purified Water, U.S.P. as the blank.

4.5.2.2 Test for correct color development. To 1.0 ml of reagent diluted as above (4.5.2.1), add 0.10 ml of 10.7 mEq ammonium standard (15 mg N/100 ml), then add 1.0 ml of diluted alkali-hypochlorite reagent. Incubate at 37° C. for 15 minutes. Add 7.0 ml of Purified Water, U.S.P. and determine absorbance of solution at 640 mu with a suitable spectrophotometer in a 1-cm cell versus a blank prepared in the same manner, except for omission of the ammonium standard.

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4.5.3 Alkali-hypochlorite reagent.

4.5.3.1 Assay for alkalinity. Add 0.100 ml of sample (concentrated reagent) to approximately 10 ml of Purified Water, U.S.P. Titrate with a microtitrator, adding first 0.400 ml of 1N Hydrochloric Acid Volumetric Solution, then 1 drop of Phenolphthalein T.S. Titrate to complete disappearance of red color. Should color reappear, add another drop of indicator and titrate to disappearance of color.

4.5.3.2 Assay for hypochlorite content. Pipet 1.0 ml sample (concentrated reagent) to approximately 10 ml Purified Water, U.S.P. Add approximately 0.5 gram KI and 0.5 ml glacial acetic acid. Titrate with 1N Sodium Thiosulfate Volumetric solution, using a microtitrator, with starch as the indicator.

4.5.4 Nitrogen standard assay.

Reagents:

Purified Water, U.S.P.

Phenol color reagent. Warm contents of bottle to liquefy, and transfer quantitatively to a 100-ml volumetric flask with the aid of Purified Water, U.S.P. and dilute to volume with Purified Water, U.S.P.

Alkali-hypochlorite reagent. Transfer quantitatively to a 100-ml volumetric flask with the aid of, and dilute to volume with Purified Water, U.S.P.

Primary Nitrogen Standard. Dry ammonium sulfate, A.C.S. at 105°to 110° C. for 2 hours. Accurately weigh 70.7 mg into a 100-ml volumetric flask and dilute to volume with the Purified Water, U.S.P. (Solution is equivalent to 15 mg percent nitrogen).

Procedure.

Pipet 1.0 ml of phenol color reagent into each of 5 test tubes. To two tubes, add 0.020 ml Primary Nitrogen Standard. To two tubes, add 0.020 ml of nitrogen standard being tested. The remaining tube is the reagent blank. To each tube, add 1.0 ml of alkali hypochlorite solution and mix. Incubate all tubes at 37° C. for 15 minutes. Add 7.0 ml of Purified Water, U.S.P. to each tube, mix, and cool to room temperature. Determine the absorbances of the four nitrogen solutions with a suitable spectrophotometer in a 1-cm cell versus the reagent blank at 640 mu.

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Calculation:

$$\frac{\text{Au sample}}{\text{As standard}} \times 15 \text{ mg \%} = \text{mg \% nitrogen in sample.}$$

where:

Au = Absorbance of nitrogen standard being tested
As = Absorbance of primary nitrogen standard.

(Average the two readings for Au and As, respectively. Au readings shall not differ from each other by more than 4.0 percent. As readings shall not differ from each other by more than 4.0 percent.)

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of Interim Federal Specification PPP-C-00186a, dated 15 May 1969, and Amendment-1, dated 27 October 1969, and as specified herein:

5.1.1 Immediate containers. Each immediate container shall comply with the classification as indicated for the reagent, as follows:

5.1.1.1 For the urease.

GROUP A	CLASS 1	TYPE a	STYLE 3	GRADE 2
CLOSURE H				

5.1.1.2 For the Phenol Color Reagent.

GROUP A	CLASS 2 - polyethylene	STYLE 1	GRADE 2
CLOSURE A - suitable liner*			SEAL A

*Closure may be urea formaldehyde resin.

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5.1.1.3 For the Alkali-Hypochlorite Reagent

GROUP A CLASS 2 - polyethylene STYLE 1 GRADE 2

CLOSURE A - without liner. (Closure may be urea formaldehyde resin).

SEAL - Shall be a plastic plug that extends well down into the neck of the bottle. The plug shall contain a small thin diaphragm which is punctured for use, permitting removal of contents when bottle is inverted and squeezed.

In addition, an outer seal shall be furnished.

5.1.1.4 For the Nitrogen Standard Solution.

GROUP A CLASS 1 TYPE e STYLE 1 GRADE 2

CLOSURE A SEAL A

5.2 Labeling. Labeling shall be in accordance with applicable regulations and shall include the information required below:

5.2.1 Immediate containers.

5.2.1.1 Urease. Each immediate container for the Urease shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) label information in accordance with commercial practice

(b) identification of the contents and its use

(c) the phrase "20 ml size" or a similar phrase

(d) the lot or control number

(e) the expiration date.

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5.2.1.2 Phenol Color Reagent. Each immediate container for the Phenol Color Reagent shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) label information in accordance with commercial practice

(b) identification of the contents, its use and toxicity

(c) the quantity of contents designated as "5 Grams"

Note: The official abbreviation "g." may be used in lieu of the word "grams."

(d) the lot or control number.

5.2.1.3 Alkali-Hypochlorite Reagent. Each immediate container for Alkali-Hypochlorite Reagent shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) label information in accordance with commercial practice

(b) identification of the contents, its use and toxicity

(c) the quantity of contents designated as "14 ml"

(d) the lot or control number.

5.2.1.4 Nitrogen Standard. Each immediate container for the Nitrogen Standard shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) label information in accordance with commercial practice

(b) identification of the contents and its use

(c) the phrase "3.5 ml" or a similar phrase

(d) the lot or control number

(e) the expiration date.

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5.2.2 Unit packages. Each unit package label shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) labeling information in accordance with commercial practice

(b) the item identification designated as "TEST KIT, NITROGEN DETERMINATION"

(c) the Federal Stock No.

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the expiration date

(g) the statement "Store between 2° - 8° C. (35° - 46° F.)"

(h) the statement: "Note: The two concentrated reagents are corrosive poisons. Use proper precautions when handling" or a similar statement.

5.2.2.1 Circulars. A brochure, circular, or other printed matter shall be included in the package which shall contain a description of the item, complete instructions for its use, and necessary precautions.

5.3 Packaging and packing.

5.3.1 Unit of issue. Each test kit, as specified, constitutes one unit of issue.

5.3.2 Unit package. The unit package shall be a rigid, transparent, hinged top plastic container of suitable size and having a snap closure. Vials shall be cushioned within the container with form molded plastic in such a manner as to minimize breakage and inadvertent opening. The plastic container shall be further packaged in a clear polyethylene bag

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of appropriate size to prevent leakage of corrosive reagents. Closure of bag shall be effected by heat-sealing.

5.3.3 Procedure code. Procedure code No. 3, as specified in Table I of PPP-C-00186a, shall apply.

5.4 Marking.

5.4.1 Intermediate package. Each intermediate package shall be marked in accordance with paragraphs 5.5.3 and 5.5.8.2 of PPP-C-00186a, except that the expiration date shall be shown. Marking shall include the legend "STORE BETWEEN 2° - 8° C. (35° - 46° F.)."

5.4.2 Exterior container. Exterior container shall be marked in accordance with paragraphs 5.5.4 and 5.5.8.2 of PPP-C-00186a, except that the expiration date shall be shown. Marking shall include the following legend:

"STORE BETWEEN 2° - 8° C. (35° - 46° F.)"

NOTE: The unrefrigerated shipping time for this item shall be not more than 3 days (72 hours).

6. NOTES

6.1 Ordering data. Purchases should exercise any desired options offered herein, and the procurement document should specify the following:

(a) Number and date of this specification
(b) the Federal Stock No.
(c) Selection of applicable levels of packaging and packing (see PPP-C-00186a).

6.2 Federal Stock No. coverage. This specification covers the following item appearing in the Federal Supply Catalog

Federal Stock No.	Item Identification	Unit
6505-165-6508	TEST KIT, NITROGEN DETERMINATION	Each

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1295

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE: Military Specification MIL-T-36888

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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